

Isomeric states in ^{273}Ds and ^{269}Hs populated in the $^{238}\text{U}(^{40}\text{Ar}, 5n)$ reaction

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The $^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ reaction was investigated across an excitation-energy range of 40 to 51 MeV using the SHANS-2 separator. Upper cross-section limits of 833, 1118, and 511 fb were established at excitation energies of 40, 45, and 50 MeV, respectively. At an excitation energy of 49.5 MeV, two decay chains of ^{273}Ds were observed, yielding a production cross section of 219_{-147}^{+292} fb. The transitions observed in these new chains exhibit distinct energies and lifetimes, corroborating previous findings and confirming the existence of an isomeric state in ^{273}Ds in addition to the ground state. The measured cross section is in excellent agreement with the previously reported value of 180_{-120}^{+440} fb obtained at the DGFRS-2 separator at 49 MeV.

I. INTRODUCTION

Superactinides are generally considered to be elements with $Z \geq 104$. According to the liquid drop model, these elements should not exist [1]. Currently, the academic community believes that the existence of such superheavy elements is related to the nuclear shell structure. The predicted limits of nuclear survival lie at the closed nuclear shells of protons $Z = 108$ and neutrons $N = 162$ [2], as well as the emergence of heavier superheavy nuclei with $Z = 114 - 126$ and $N = 184$ [3]. As an odd-neutron nucleus with a single particle on the $N = 162$ shell, ^{273}Ds and its decay product ^{269}Hs , which is an odd-neutron nucleus with a hole below the $N = 161$ shell, are of great importance for gaining deeper insights into the $N = 162$ shell.

^{273}Ds was first unambiguously observed as a decay product in cold fusion experiments at the SHIP facility at GSI [4]. It is worth noting that in the cold fusion reaction $^{208}\text{Pb}(^{70}\text{Zn}, n)^{277}\text{Cn}$ conducted at GSI [4, 5] and RIKEN [6, 7], the lifetime of ^{273}Ds as a decay product is on the order of 0.1 ms. Last year, in the

$^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ experiment conducted by Yu. Ts. Oganessian *et al.* [8] at the DGFRS-2 separator [9], two decay chains were discovered where ^{273}Ds was produced directly. These were identified using the genetic correlation method [10]. One chain is consistent with historical data, while the other was assigned to an isomeric state of ^{273}Ds . Unfortunately, both chains are incomplete, and little is known about the decay information of ^{269}Hs .

Subsequently, Z. Wang *et al.* performed theoretical calculations on the decay pathways of these two chains using the density-dependent cluster model (DDCM) and the unified decay formula (UDF) [11], proposing minor modifications to these two chains. The present work was carried out at SHANS-2 [12], where two additional distinct decay pathways were observed using the same reaction. The pathway information is complete and consistent with historical data through the genetic correlation method [10], which fully and independently confirms the authenticity of the two states of ^{273}Ds ($^{273}\text{Ds}^a$ and $^{273}\text{Ds}^b$) observed by Yu. Ts. Oganessian *et al.* [8].

The experiment is described below in Table I, and the experimental data are presented.

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TABLE I. Experimental parameters and cross sections for the $^{238}\text{U} + ^{40}\text{Ar}$ reaction.

Reaction	Target thickness (mg/cm ²)	E_{lab} (MeV)	E^* (MeV)	Beam dose $\times 10^{19}$	No. of chains 4n/5n	σ_{4n} (pb)	σ_{5n} (pb)
$^{238}\text{U} + ^{40}\text{Ar}$	0.47 / 0.60	212.7	50.6	1.95 / 1.45	0/2	< 0.43	$0.15^{+0.20}_{-0.10}$
	0.47	207.4	46.2	0.82	0/0	< 0.9	< 0.9
	0.60	204.0	43.6	2.04	0/0	< 0.3	< 0.3
	0.60	214.3	50.9	1.527	0/0	< 0.4	< 0.4

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